

Vector $\rightarrow$ object defined with a magnitude and direction

Formally in mathematics:-

They are elements of a vector space: a collection of objects that is closed under an addition rule and a rule for multiplication by scalars.

In 1-dimension

Vector is represented as a 1-dimensional array of numbers; referred to as components and is displayed either in column or row form.

One should think vectors as a set^{of number}, which typically represent coordinates within a n -dimensional space, where n is the number of dimensions.

Most simple ~~representation~~ things come to our mind when we think of vector

$\vec{v}$ or $\mathbf{v}$

Simple vector $\rightarrow$ arrow in a vector space, with origin, direction and magnitude (length)

Vectors in $\mathbb{R}^n$

$\mathbb{R} \rightarrow$ real number
 $n \rightarrow$ dimension.

L6-1b

$$V = [v_1, v_2, \dots, v_n]$$

$v_i \in \mathbb{R}^n$
real vector of n dimensions

$$V = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$

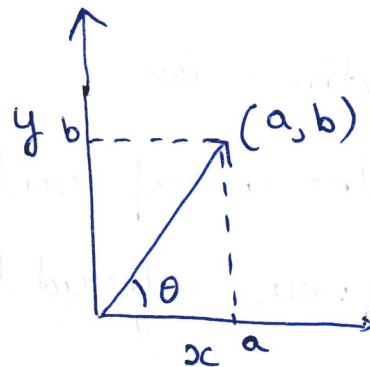
Let's take $n=2$.

which gives us 2-D.

$$V = (a, b)$$

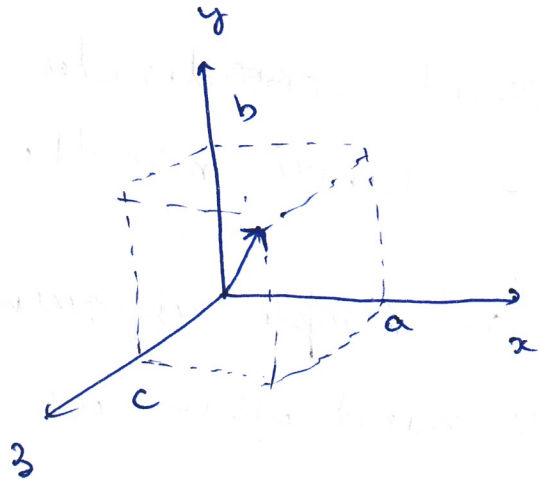
length is $|V|$

$\theta \rightarrow$ angle of vector with x -axis



$n=3$

$$V = (a, b, c)$$



Visualizing after this become almost impossible for human as we have not experienced the higher dimension.

L6-2a

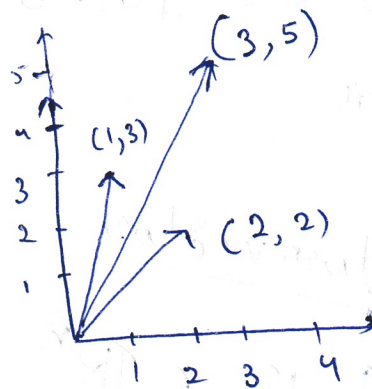
Vector Algebra

Addition
Subtraction
 $\mathbb{R}^2$

$$v_1 = (1, 3) \text{ and } v_2 = (2, 2)$$

$$v_1 + v_2 = (3, 5)$$

$$v_1 - v_2 = (-1, 1)$$



$$\mathbb{R}^n \quad v_1 = (x_1, x_2, x_3, \dots, x_n) \quad v_2 = (y_1, y_2, y_3, \dots, y_n)$$

$$v_1 \pm v_2 = (x_1 \pm y_1, x_2 \pm y_2, x_3 \pm y_3, \dots, x_n \pm y_n)$$

Dot Product

$$v_1 = (x_1, x_2, \dots, x_n) \quad v_2 = (y_1, y_2, y_3, \dots, y_n)$$

$$\begin{aligned} v_1 \cdot v_2 &= x_1 y_1 + x_2 y_2 + x_3 y_3 + \dots + x_n y_n \\ &= \sum_{i=1}^n x_i y_i \end{aligned}$$

$$\text{In } \mathbb{R}^2 \quad v_1 = (1, 3) + v_2 = (2, 2) \quad v_1 \cdot v_2 = 2 + 6 = 8$$

Length $v \text{ in } \mathbb{R}^n$

$$v = (x_1, x_2, \dots, x_n)$$

$$\begin{aligned} |\vec{v}| &= \sqrt{\vec{v} \cdot \vec{v}} = \sqrt{x_1^2 + x_2^2 + x_3^2 + \dots + x_n^2} \\ &= \sqrt{\sum_{i=1}^n x_i^2} \end{aligned}$$

$$|v_1| = \sqrt{1+9} = \sqrt{10}$$

$$|v_2| = \sqrt{4+4} = \sqrt{8}$$

If we found that $|\vec{v}| = 0$ then we call it zero vector.

Angle b/w two vectors

$$\vec{v}_1, \vec{v}_2 \in \mathbb{R}^n$$

$$\cos \theta = \frac{\vec{v}_1 \cdot \vec{v}_2}{|\vec{v}_1| |\vec{v}_2|}$$

Scalar multiplication

$$\vec{v} \in \mathbb{R}^n \quad \vec{v} = (x_1, x_2, x_3, \dots, x_n)$$

$$a \cdot \vec{v}$$

a is scalar

$$a \cdot \vec{v} = (ax_1, ax_2, ax_3, \dots, ax_n)$$

$$5 \cdot \vec{v}_1 = (1, 3)$$

$$5 \cdot \vec{v}_1 = (5, 15)$$

It simply increase magnitude

$$|5 \cdot \vec{v}_1| = \sqrt{25 + 225} = \sqrt{250} = 5\sqrt{10}$$

$$|\vec{v}_1| = \sqrt{10}$$

$$\boxed{|5 \cdot \vec{v}_1| = 5 |\vec{v}_1|}$$

Dot product

It encodes a lot of geometry.

Consider two vectors

$$x, y \in \mathbb{R}^d$$

$$x^T y = x \cdot y = \langle x, y \rangle = [x_1, x_2, \dots, x_d] \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{bmatrix} = \sum_{i=1}^d x_i y_i$$

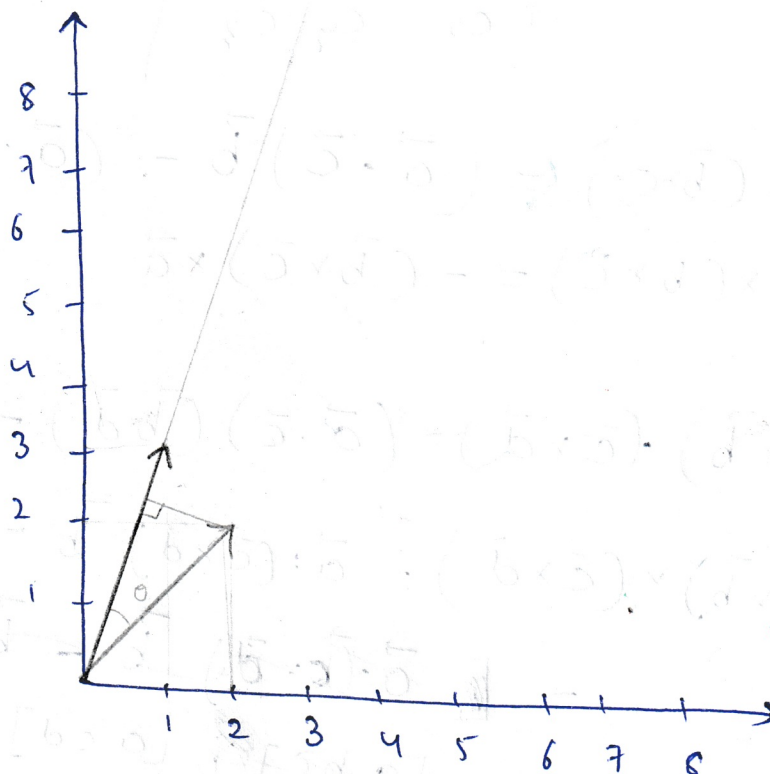
$$\langle x, y \rangle = \text{length}(x) \cdot \text{length}(y) \cdot \cos(\theta)$$

$$\text{Let } x = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad y = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$\langle x, y \rangle = x \cdot y = [1 \ 3] \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2 + 6 = 8$$

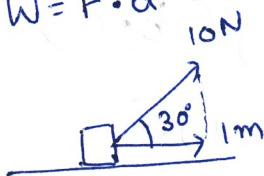
$$x = \sqrt{1+9} = \sqrt{10}$$

$$y = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$$



Work done

$$W = F \cdot d$$



$$10 \cdot \frac{\sqrt{3}}{2} \cdot 1$$

~~$\cos 30^\circ$~~

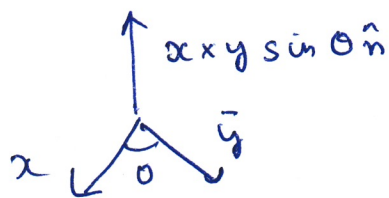
Work done

$$5\sqrt{3} \text{ N/m}$$

Gross Product

$$\vec{x} \times \vec{y} = xy \sin \theta \hat{n}$$

θ is angle b/w $\vec{x}$ & $\vec{y}$ & $\hat{n}$ is a vector along $\vec{x} \times \vec{y}$



$\hat{n}$ is determined by right-hand screw rule

$$\vec{x} \cdot \vec{x} \times \vec{y} \neq \vec{y} \times \vec{x}$$

→ Condition for two collinear vectors

Two vectors are collinear if $\theta = 0$ or π i.e. $\sin \theta = 0$

$$\vec{x} \times \vec{y} = 0$$

→ Product of 3 vectors

Scalar $\vec{a} \cdot (\vec{b} \times \vec{c}) = [\vec{a} \vec{b} \vec{c}]$

$$= \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}$$

Vector $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$

$$\Rightarrow \vec{a} \times (\vec{b} \times \vec{c}) = -(\vec{b} \times \vec{c}) \times \vec{a}$$

4 vector scalar $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c})$

vector $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{a} \cdot (\vec{c} \times \vec{d}) \vec{b} - \vec{b} \cdot (\vec{c} \times \vec{d}) \vec{a}$

$$= \cancel{[\vec{a} \vec{b} \vec{c} \vec{d}]} \vec{b} - \cancel{[\vec{b} \vec{c} \vec{d} \vec{a}]} \vec{a}$$

$$= [\vec{a} \vec{b} \vec{c} \vec{d}] \vec{b} - [\vec{b} \vec{c} \vec{d} \vec{a}] \vec{a}$$

Now we know what vectors are.
 Linear algebra simply and beautifully focuses on these vectors.

Linear algebra plays an important role in machine learning and general mathematics.

Ex 88; Matrices

A vector $v = (v_1, v_2, \dots, v_n)$ is equivalent to a point in $\mathbb{R}^n$ space.

By default vector is a column of n numbers

$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

but in some cases we assume it as row

$$v^T = [v_1 \ v_2 \ \dots \ v_n]$$

$n \times 1$ - matrix

An $n \times d$ matrix A is then an ordered set of n row vectors $a_1, a_2, \dots, a_n$.

$$A = [a_1; a_2; a_3; \dots; a_n] =$$

$$\begin{bmatrix} \text{---} a_1 \text{---} \\ \text{---} a_2 \text{---} \\ \vdots \\ \text{---} a_n \text{---} \end{bmatrix}$$

$$= \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1d} \\ A_{21} & A_{22} & \dots & A_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nd} \end{bmatrix}$$

here $a_i = [A_{i1}, A_{i2}, A_{i3}, \dots, A_{id}]$

A_{ij} is the element of n in the i th row and j th column.

One can write $A \in \mathbb{R}^{n \times d}$

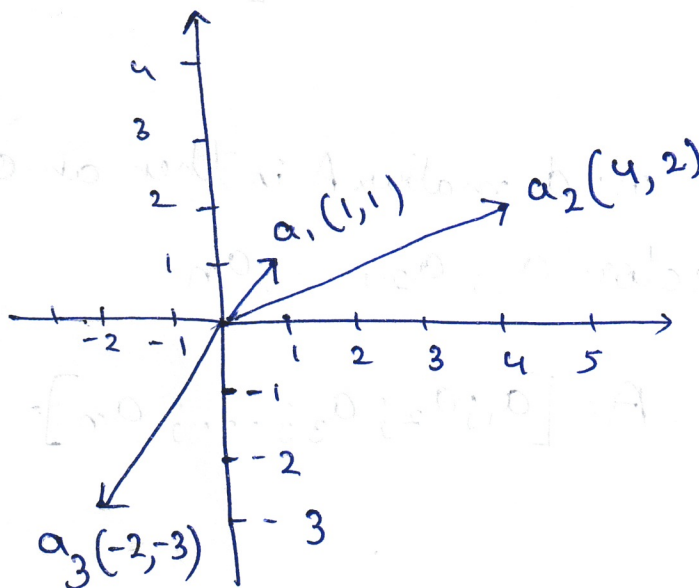
How to visualize?

The most easiest way is to ^{first} imagine d vectors $\vec{v}$ as points in $\mathbb{R}^d$.

Similarly, it is convenient to imagine $n \times d$ n A by each of its rows $a_1, \dots, a_n$ as each points in $\mathbb{R}^d$ space.

3×2 n

$$A = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & 2 \\ -2 & -3 \end{bmatrix}$$



L6-4a

All of us has heard of the transpose.

Transpose operation $(\cdot)^T$ simply reverse the role of the rows and column.

$$A = \begin{bmatrix} -a_1 - \\ -a_2 - \\ \vdots \\ -a_n - \end{bmatrix} \quad A^T = \begin{bmatrix} -a_1 - \\ -a_2 - \\ \vdots \\ -a_n - \end{bmatrix}^T = \begin{bmatrix} | & | & \dots & | \\ a_1 & a_2 & \dots & a_n \\ | & | & \dots & | \end{bmatrix}$$

$$A^T = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1d} \\ A_{21} & A_{22} & \dots & A_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nd} \end{bmatrix}^T = \begin{bmatrix} A_{11} & A_{21} & \dots & A_{n1} \\ A_{12} & A_{22} & \dots & A_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1d} & A_{2d} & \dots & A_{nd} \end{bmatrix}$$

Let see how these matrices behave.

Addition

As we see in the vectors, similarly we can add two matrices only if they have same dimension.

$$A, B \in \mathbb{R}^{n \times d}$$

then $C = A + B$ where $C_{ij} = A_{ij} + B_{ij}$ for all i, j

Multiplication

Multiplication only require alignment along the one dimension.

For two matrices $A \in \mathbb{R}^{n \times d}$; $B \in \mathbb{R}^{d \times m}$

We obtain $\rightarrow$ $C = A \cdot B \in \mathbb{R}^{n \times m}$

where

$$C_{ij} = \sum_{k=1}^d A_{i,k} B_{k,j}$$

L 6-4b

Necessary condition for multiplication of two matrices A and B is that:-

no. of columns of A = no. of rows of B.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3} \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix}_{3 \times 4}$$

||| ≡

$$C_{11} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31}$$

$$C_{12} = (a_{11} \ a_{12} \ a_{13}) \begin{pmatrix} b_{12} \\ b_{22} \\ b_{32} \end{pmatrix}$$

$$C_{13} = (a_{11} \ a_{12} \ a_{13}) \begin{pmatrix} b_{13} \\ b_{23} \\ b_{33} \end{pmatrix}$$

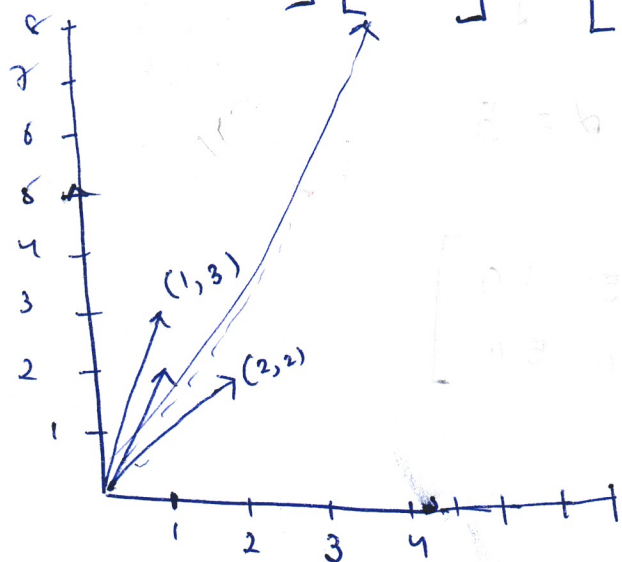
$$C_{14} = (a_{11} \ a_{12} \ a_{13}) \begin{pmatrix} b_{14} \\ b_{24} \\ b_{34} \end{pmatrix}$$

$$C_{21} = (a_{21} \ a_{22} \ a_{23}) \begin{pmatrix} b_{11} \\ b_{21} \\ b_{31} \end{pmatrix}$$

and so on

L6-5a

$$\begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2+2 \\ 2+6 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \end{bmatrix}$$



$$\begin{matrix} 2 \times 2 \\ 2 \times 1 \\ 2 \times 1 \end{matrix}$$

Not straight forward but one can simply imagine matrix multiplication like some transformation of the space, vectors.

→ Def Properties of matrix multiplication

① Associative

$$(AB)C = A(BC)$$

② Distributive $A(B+C) = AB + AC$

second distributive $(A+B)C = AC + BC$

③ Not commutative

$$AB \neq BA$$

We can also multiply $\mathbb{R}$ A by a scalar

L6-5b

α -scalar

αA is simply αA_{ij}

$$A = \begin{bmatrix} 7 & 2 \\ 6 & 10 \end{bmatrix} \quad \alpha = 5$$

$$\alpha A = \begin{bmatrix} 35 & 10 \\ 30 & 50 \end{bmatrix}$$

$\alpha \in \mathbb{R}$ or $\mathbb{C}$ complex

Remarks

→ If $AB = 0$; then you should not expect
 $A = 0$ or $B = 0$ they can be non-zero

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

→ If $AB = AC$, B not necessarily $= C$.

L6-6a

Few of matrices

Diagonal $n \times n$ square $n \times n$
off diagonal $n \times n$ elements are all
zero

$$\begin{bmatrix} 7 & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 10 \end{bmatrix}$$

$i = j \rightarrow$ non-zero
 $i \neq j \rightarrow$ zero

Triangular $n \times n$

Square $n \times n$ whose elements above
or below the main diagonal are
zero

$$\begin{bmatrix} 7 & 8 & 9 \\ 0 & 8 & 8 \\ 0 & 0 & 9 \end{bmatrix} \text{ upper}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 4 & 0 \\ 3 & 5 & 6 \end{bmatrix} \text{ lower}$$

Identity $n \times n$

A diagonal $n \times n$ whose all diagonal entries are
all 1

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Square $n \times n$

Two matrices are said to be equal if their dimensions are equal and all corresponding elements too.

L6-6b

Transpose of a matrix

Involve transforming its row into column and its column into rows.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

Properties of transposed matrix

$$(A+B)^T = A^T + B^T$$

$$(AB)^T = B^T A^T$$

$$(\alpha A)^T = \alpha A^T$$

$$(A^T)^T = A$$

Determinant of a matrix

Every square $n \times n$ has a determinant, which is a single numerical value used when calculating the inverse or when solving system of linear $= n$.

denoted as $|A|$ or $\det(A)$

2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad |A| = ad - bc$$

3×3 matrix

$$A = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Inverse of a matrix

If one has a scalar a , then its inverse is simply $1/a$

$$a \cdot \frac{1}{a} = 1$$

Division of matrices is not defined

~~since~~ as we know - that

Even if $AB = AC$, we can't claim $B = C$.

One use matrix inversion.

The inverse of matrix A (square $n \times n$), if exist

is a unique $n \times n$ A^{-1} where I is identity $n \times n$

$$A A^{-1} = A^{-1} A = I$$

For this A should be a non-singular $n \times n$.

Singular $n \times n$ means $\det(A) = 0$.

For A^{-1} , we should have $\det(A) \neq 0$.

Please note A^{-1} is not $1/A$

Inverse of 2×2 $n \times n$

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$A^{-1} = \frac{1}{|A|} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

For smaller $n \times n$ it is easier

Cofactors

The inverse of larger $n \times n$ is computed by using cofactor.

Let A be a square $n \times n$.

We define Minor (denoted as M_{ij})

Minor is the entry in the i^{th} row & j^{th} column & is the determinant of the submatrix formed by deleting the i^{th} row & j^{th} column.

The (i, j) cofactor is obtained by multiplying the Minor M_{ij} by $(-1)^{i+j}$

$$C = \begin{pmatrix} +M_{11} & -M_{12} & +M_{13} \\ -M_{21} & +M_{22} & -M_{23} \\ +M_{31} & -M_{32} & +M_{33} \end{pmatrix}$$

2nd row & first col

$$i=2, j=1 \quad \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

$$M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

$$A^{-1} = \frac{1}{|A|} C^T$$

$$|A|$$

$$= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

$$M_{23} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

$$M_{32} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$$

and so on.

Adjoint of A :- defined as the transpose of matrix formed by cofactor of element of A .

$$A^{-1} = \frac{\text{adj } A}{|A|}$$

Find the inverse of A

$$\begin{bmatrix} 1 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & -1 \end{bmatrix}$$

$$|A| = 1(-1-4) - (-1)(1-6) + 3(-2-3) = -25$$

Cofactors of first row.

$\begin{pmatrix} 1 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix}$ $(-1) - (4) = -5$ $(-1)^{1+1}(-5) = -5$	$\begin{pmatrix} 1 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix}$ $(1-6) = -5$ $(-1)^{1+2}(-5) = 5$	$\begin{pmatrix} 1 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix}$ $(-1)^{1+3}(-2-5) = -5$
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Second row

$\begin{pmatrix} 1 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix}$ $(-1)^{2+1}(1-6) = 5$	$\begin{pmatrix} 1 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix}$ $(-1)^{2+2}(-1-9) = -10$	$\begin{pmatrix} 1 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix}$ $(-1)^{2+3}(2+3) = -5$
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Third row

$\begin{pmatrix} 1 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix}$ $(-1)^{3+1}(-2-3) = -5$	$\begin{pmatrix} 1 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix}$ $(-1)^{3+2}(2+3) = -5$	$\begin{pmatrix} 1 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & -1 \end{pmatrix}$ $(-1)^{3+3}(1-1) = 0$
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$$C = \begin{pmatrix} -5 & 5 & -5 \\ 5 & -10 & -5 \\ -5 & -5 & 0 \end{pmatrix}$$

$$\text{adj } A = C^T = \begin{bmatrix} -5 & 5 & -5 \\ 5 & -10 & -5 \\ -5 & -5 & 0 \end{bmatrix}$$

$$A^{-1} = \frac{\text{adj } A}{|A|} = -\frac{1}{25} \begin{bmatrix} -5 & 5 & -5 \\ 5 & -10 & -5 \\ -5 & -5 & 0 \end{bmatrix} = \begin{bmatrix} 1/5 & -1/5 & 1/5 \\ -1/5 & 2/5 & 1/5 \\ 1/5 & 1/5 & 0 \end{bmatrix}$$

L6-9a

The formula for the inverse

matrix A^{-1} is

$$A^{-1} = \frac{1}{|A|} C^T$$

Properties of Inverse of a matrix

$$(AB)^{-1} = B^{-1} A^{-1}$$

$$(A^{-1})^{-1} = A$$

$$(A^T)^{-1} = (A^{-1})^T$$

$$(kA)^{-1} = \frac{1}{k} (A^{-1})$$

- Square matrix having inverse is called Non Singular matrix.
- matrix with no inverse called Singular matrix.

Linear Combination of vectors

Consider that you have a set of vectors

$$S = (v_1, v_2, \dots, v_g)$$

Then you can make a new vector using the same vector space

where $d_1, d_2, \dots, d_g$ are scalars

$$v = d_1 v_1 + d_2 v_2 + \dots + d_g v_g$$

This combination is called or known as

linear combination of $v_1, v_2, \dots, v_g$

Simple example

L6-9b

$$v_1 = (2, 3, -1), v_2 = (-1, 3, 6); v_3 = (1, 0, 5)$$

$$v_4 = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$$

$$= \alpha_1 (2, 3, -1) + \alpha_2 (-1, 3, 6) + \alpha_3 (1, 0, 5)$$

$$= (2\alpha_1 - \alpha_2 + \alpha_3, 3\alpha_1 + 3\alpha_2, -\alpha_1 + 6\alpha_2 + 5\alpha_3)$$

$$\text{if } \alpha_1 = 2, \alpha_2 = 3 + \alpha_3 = 6$$

$$v_4 = (7, 15, 46)$$

===== This way one can get other

vectors!

Let say: a set of vectors $S = \{v_1, v_2, \dots, v_g\}$ and it is called linearly independent of the vector equation.

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_g v_g = 0$$

holds only if $\alpha_1 = \alpha_2 = \alpha_3 = \dots = \alpha_g = 0$.

Else the set S is linearly dependent.

$$\text{E.g. } S \quad v_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \quad v_2 = \begin{pmatrix} 3 \\ 9 \end{pmatrix}$$

$$\text{The l.c. } -3v_1 + v_2 = 0$$

$$-3 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ 9 \end{pmatrix} = 0$$

There v_1 & v_2 are linearly dependent

L6-10a

Linearly independent vector

Simple example

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\alpha_1 v_1 + \alpha_2 v_2$$

$$= \alpha_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

As a consequence they
are linearly independent

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

$$v_2 = \frac{3}{2} v_3$$

$$2v_2 - 3v_3 = 0$$

$$\alpha_1 = 0, \alpha_2 = 2, \alpha_3 = -3 \text{ gives}$$

0.

There exists a linear combination of 3 vectors such
that coefficients of combination are not all
equal to zero

$$S = \{(1, -1, 0) \quad (1, 0, 1) \quad (0, 1, 1)\}$$

L 6-10b

↓
L.D. set of vector $\alpha_1=1, \alpha_2=-1, \alpha_3=1$

Remark

- ① In $\mathbb{R}^n$, a set of more than n vectors is linearly dependent
- ② Any set of vectors containing zero vector is also linearly dependent (trivial)

⊥ orthogonal

Orthogonal Vectors (⊥) →

We say that a set of vectors $(v_1, v_2, v_3, \dots, v_n)$ are orthogonal.

when they are mutually or pairwise orthogonal. which means their dot product is

zero. $\vec{v}_i \cdot \vec{v}_j = 0$ for $i \neq j$

→ An orthogonal set of non-zero vectors is linearly independent

L6-11a

Example

$$v_1, v_2, v_3 \in \mathbb{R}^3$$

$$\left\{ \overset{v_1}{(2, 0, -2)}, \overset{v_2}{(2, -\sqrt{8}, 2)}, \overset{v_3}{(2, \sqrt{8}, 2)} \right\}$$

$$v_1 \cdot v_2 = 4 + 0 - 4 = 0$$

$$v_1 \cdot v_3 = 4 + 0 - 4 = 0$$

$$v_2 \cdot v_3 = 4 - 8 + 4 = 0$$

They are pairwise orthogonal

Ortho normal

A set of orthogonal vector is called ~~ortho~~ orthonormal if each vector has length 1

$$v_1 =$$

$$(1/\sqrt{2}, 0, -1/\sqrt{2}); (1/2, -\sqrt{2}/2, 1/2); (1/2, \sqrt{2}/2, 1/2)$$

They are orthonormal vector set

* Let say if $\alpha \vec{v}_1 + \beta \vec{v}_2 = \vec{0}$;

and α, β are not all zero, then we know that these two vectors are linearly dependent.

Two linearly dependent vectors are parallel $(\vec{v}_1 \parallel \vec{v}_2)$ to each other.

Similarly, any 3 linearly dependent vectors will be in same plane.



Basis:-

In n-dimensional space one can choose a number of sets, each set containing n vectors $(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n)$ such that the vectors of each set are linearly independent and span the space.

Any set of n such vectors in n-dimensional space is said to form a set of basis vectors or to provide a basis.

L6-12a

Expansion Theorem

If the vectors $v_1, v_2, \dots, v_n$ form a basis, then any vector α can be expressed as a linear combination of the vectors v_i .

$$\begin{aligned}\alpha &= \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \dots + \alpha_n v_n \\ &= \sum_i^n \alpha_i v_i\end{aligned}$$

Simplest example

$$v_1 = \{1, 0, 0, \dots, 0\}$$

$$v_2 = \{0, 1, 0, \dots, 0\}$$

$$v_3 = \{0, 0, 1, \dots, 0\}$$

$\vdots$

$$v_n = \{0, 0, 0, \dots, 1\}$$

$$\alpha = \alpha_1 v_1, \alpha = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \dots + \alpha_n v_n$$